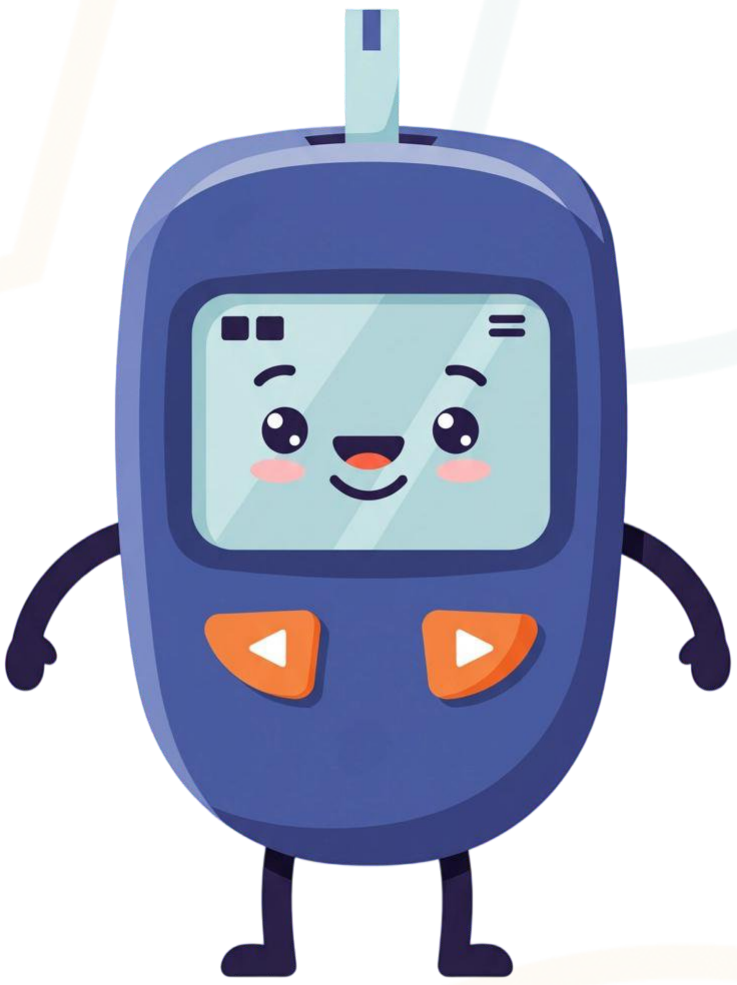
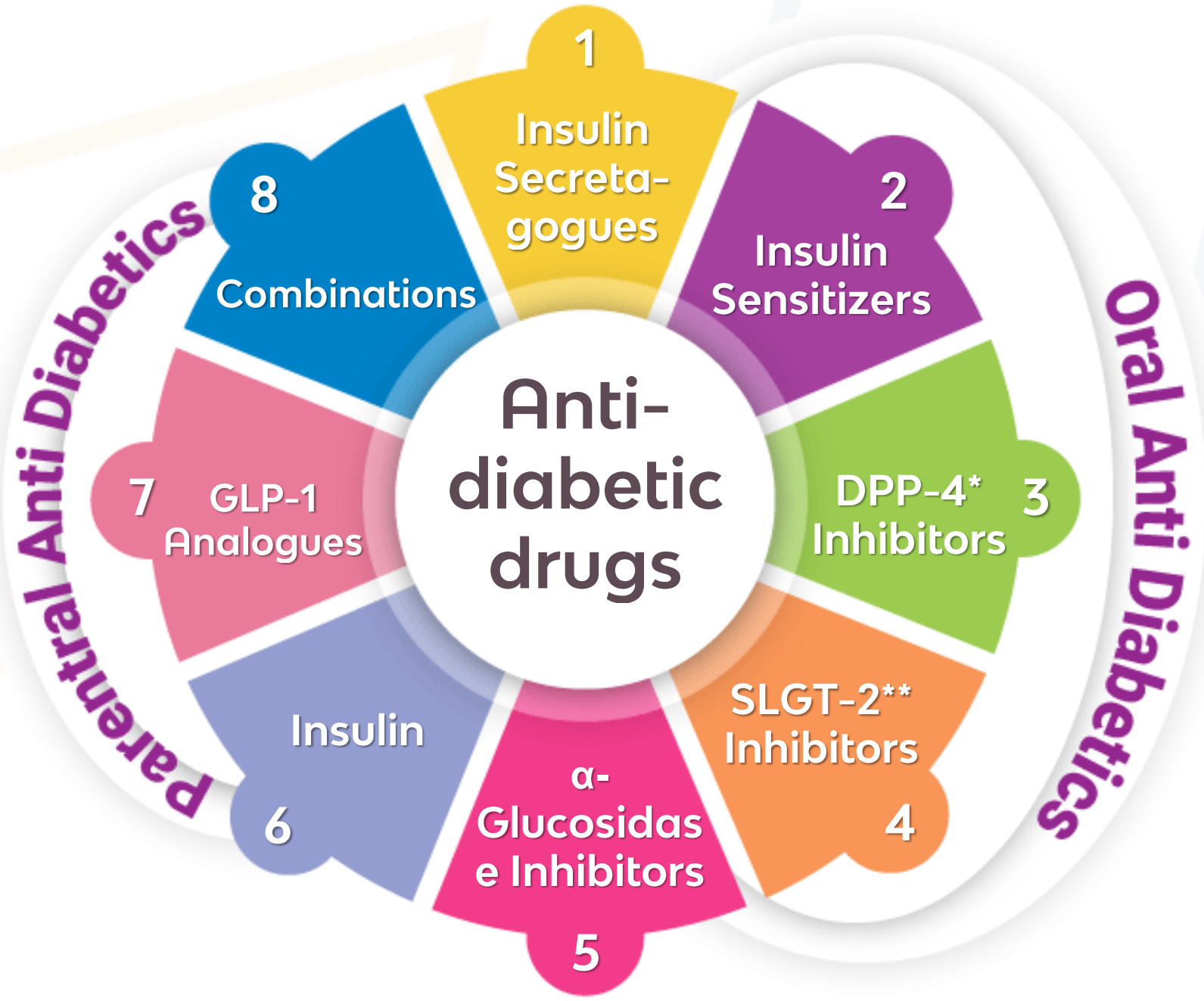
Diabetes Part 1



Talent Management Department

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Diabetes drugs classification



*** (DPP-4) ➡** Dipeptidyl peptidase-4
**** SLGT-2 ➡** Sodium-dependent glucose transporter-2

Today we will learn about

01 DPP-4 Inhibitors

02 Insulin Secretagogues

03 Insulin Sensitizers

04 SLGT-2 Inhibitors

05 α-Glucosidase Inhibitors

Diabetes Overview

Diabetes mellitus is a chronic metabolic disorder characterized by high blood glucose levels (hyperglycemia) due to insulin deficiency, insulin resistance, or both.

1. Type 1 Diabetes (T1DM): ✅

- Autoimmune destruction of beta cells → No insulin production.
- Requires lifelong insulin therapy.
- Common in children and young adults.

2. Type 2 Diabetes (T2DM) ✅

- Caused by insulin resistance and relative insulin deficiency.
- Strongly linked to obesity and lifestyle.
- Managed with lifestyle changes, oral drugs, and sometimes insulin.

Goals of Therapy: 🎯

- Control blood glucose
- Prevent complications (neuropathy, nephropathy, retinopathy, cardiovascular disease)
- Improve quality of life.

Insulin Secretagogues

Used in Type 2 Diabetes Mellitus to increase insulin secretion from pancreatic beta cells.

1. Sulfonylureas ✅

- They Stimulate insulin release.
- They have a risk of causing hypoglycemia.
- Not suitable for people sensitive to sulfa.

2. Meglitinides ✅

- They are Similar to sulfonylureas but faster onset of action and shorter duration.
- Has Lower risk of hypoglycemia than sulfonylureas.
- Can be used safely in sulfa allergic patients.

Insulin Sensitizers

They Improve body's response to insulin.

1. Biguanides ✅

- Increase insulin sensitivity & decrease liver glucose production (gluconeogenesis).
- Have weight loss effect.
- May cause GI upset.

2. Thiazolidinediones ✅

- Increase Insulin sensitivity.
- Should be used with caution in patients with cardiovascular diseases.

1 Insulin Secretagogues

Sulfonylureas

- **AMARYL M®|** *Glimepiride & Metformin*
- **DUETACT®|** *Glimepiride & Pioglitazone*
- **AMARYL®, GLIM® |** *Glimepiride*
- **GLIBIL®, DAONIL®|** *Glyburide*
- **DIAOPTIM®, CLAZ®, DIAMICRON®|** *Gliclazide*

Meglitinides

- **NOVONORM® |** *Repaglinide*

2 Insulin Sensitizers

Biguanides

- METFOR®, GLUCARE®, FORMIT®, OMFORMIN®, GLUCOPHAGE®| *Metformin*

Thiazolidinediones

- GLADOS®, ACTOS®| *Pioglitazone*
- ACTOSMET®| *Metformin & Pioglitazone*

DPP-4 Inhibitors

DPP-4 inhibitors are oral anti-diabetic drugs used in type 2 diabetes mellitus to help lower blood glucose by enhancing the body's natural incretin system so more insulin releasees and less glucagon.

- They are Often used in combination with metformin or other oral agents.

SGLT2 Inhibitors

SGLT2 inhibitors are oral anti-diabetic drugs used in type 2 diabetes mellitus. They help lower blood glucose by increasing glucose excretion in urine.

- They have Cardiovascular & renal protection effects.
- They have Urinary tract infections & Genital infections risks.
- Should be used with caution in renal impairment patients.

3 Dipeptidyl peptidase-4 (DPP-4) Inhibitors

- **MELIGLPTIN®, JANUVIA®** | *Sitagliptin*
- **JALRA®, GALVUS®** | *Vildagliptin*
- **VIPIDIA®** | *Alogliptin*
- **ONGLYZA®** | *Saxagliptin*

- **JANUMET®** | *Sitagliptin & Metformin*
- **GALVUS MET®** | *Vildagliptin & Metformin*
- **KOMBIGLYZE®** | *Saxagliptin & Metformin*

4 Sodium Glucose Transporter 2 (SLGT2) Inhibitors

- **INVOKANA®** | *Canagliflozin*
- **DIVINUS®, DAPXIGA®, FORXIGA®** | *Dapagliflozin*
- **JARDIANCE®** | *Empagliflozin*

- **ENFORZA®** | *Saxagliptin & Dapagliflozin*
- **SYNJARDI®** | *Empagliflozin & Metformin*
- **Xigduo®** | *Dapagliflozin & Metformin*

5 α-Glucosidase Inhibitors

- **GLUCOBAY®** | *Acarbose*

Alpha Glucosidase inhibitors Acts by Delaying digestion and absorption of carbohydrates in the small intestine, thereby slowing after-meal glucose absorption.

الآن حان دور الأنشطة
اضغط على الزر للبدء



ابدأ النشاط